

Accidental Symmetry by Operator Dimension: Baryon, Lepton, and Their Remnants

Fixed-target closure of accidental charge in the Standard Model Symmetry Arc

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Principal result

For the fixed Standard Model target family declared in this paper, accidental baryon and lepton actions, their electroweak-anomaly and topology qualifications, neutrino-sector deformations, dimension-graded violating operators, and the resulting continuous or finite remnants admit a single typed classification. The finite construction is organized by 13 registered target schemas—twelve fixed schemas and the effect-set-indexed remnant schema $T(S)$ —together with 34 canonical typed scientific results arranged in seven scientific families and 12 registered effect sets. Of those 34 results, 31 are target-local classes under target-preserving equivalence, one is a cross-target constraint, one is a target-family synthesis, and one is an integrated closure result. Baryon number and the family lepton phases are derived from the renormalizable grammar rather than imposed as primitive laws; electroweak topology has transfer vector $(3, 1, 1, 1)$ in the physical (B, L_e, L_μ, L_τ) basis; Majorana and Weinberg support supply $\Delta L = 2$ breaking; and the declared higher-dimensional sectors supply the corresponding baryon/lepton violation lattices. Smith reduction is interpreted physically only after the resulting generators are mapped back to the elementary fields and overlaps with color center and fermion parity are removed. The two target-distinct effect sets each leave one order-three operator-level occurrence; they are not simultaneous factors of a single target and remain unpromoted unless and until the global-form and discrete-anomaly conditions assigned to SYM-9 are supplied.

Abstract

Accidental symmetries of the Standard Model are consequences of field content, operator support, flavor structure, and quantum sector rather than primitive prohibitions. Their status therefore changes when any of those ingredients changes. This paper develops SYM-7 of the Standard Model Symmetry Arc: a fixed-target classification of baryon number, total and family lepton numbers, $B \pm L$, and the continuous or finite remnants produced by specified symmetry-violating effects. The construction imports the symmetry-status calculus and faithful-action discipline of the earlier arc, the electroweak anomaly/topology data of SYM-3, the discrete-action grammar of SYM-4, and the flavor/neutrino and chiral-normalization interfaces supplied upstream. It then keeps four logically distinct levels separate: a symmetry action on the local algebra, the behavior of a current, perturbative selection rules in the trivial electroweak sector, and transfer between nontrivial topological sectors. For the minimal renormalizable theory, baryon number and three independent family lepton phases follow from exhaustive monomial neutrality. Electroweak topology has transfer

vector $(\Delta B, \Delta L_e, \Delta L_\mu, \Delta L_\tau) = (3, 1, 1, 1)Q_{\text{EW}}$, leaving $B - L$ and two family differences in the integer null lattice. Distinct neutrino targets separate the massless-neutrino theory, a Dirac ν_R theory, renormalizable Majorana breaking, and the Weinberg-operator theory. Source-qualified operator sectors at dimensions 3, 5, 6, 7, and a selected complete $\Delta B = 2$ dimension-nine sector are assigned exact charge signatures and coefficient support, while the dimension-18 electroweak vertex remains a topological effect rather than an ordinary Wilson operator. For twelve fixed effect sets, the resulting violation lattices are reduced by Smith normal form, mapped back to physical charge generators, and quotiented only after pre-existing overlaps are identified on the elementary fields. The final 34-result typed atlas contains 31 target-local equivalence classes together with one cross-target constraint, one target-family synthesis, and one integrated closure result. It makes the finite SYM-7 population visible in reader-facing form and defines explicit handoffs to the global-form/discrete-anomaly, renormalization-group, and whole-arc successor papers.

Keywords: accidental symmetry; baryon number; lepton number; electroweak anomaly; SMEFT; Smith normal form; discrete remnants; $B - L$; AASC.

Exact scope. The paper concerns structural statements at fixed field content, operator support, flavor stratum, background content, and quantum sector. It does not infer a symmetry from an observed small rate, identify an allowed operator with a nonzero coefficient, promote a local anomaly cancellation to a global gauging theorem, or infer ultraviolet physics from an effective remnant. Global-form completion, line/monopole sectors, discrete anomalies, non-invertible extensions, numerical Wilson-coefficient evolution, and whole-arc exhaustion are retained as explicit successor burdens rather than silently absorbed into the present target.

Read this first: arc position and proof spine

SYM-7 is not an isolated catalogue of familiar conservation laws. It is the accidental-charge stage of a dependency-ordered symmetry program. SYM-0 fixes the distinction between physical action, diagnostic, obstruction, redescription, and derived standing; SYM-1 fixes faithful gauge action and quotient discipline; SYM-2 fixes the spacetime domain anchor; SYM-3 supplies the anomaly and electroweak-topology interface; SYM-4 supplies the discrete-action grammar and the rule that a remnant is not exact merely because a continuous phase has been broken; SYM-5 supplies the flavor and neutrino target bifurcations; and SYM-6 supplies the baryon/chiral normalization interface used here. These are imported as upstream results, not re-proved under new names. [1–7]

Within AASC, a determinate symmetry claim is already made on a fixed identity-bearing target and therefore already carries the Admissibility–Standing–Reference–Irreversibility kernel. That logical dependency is imported from the foundations and SYM-0. The present paper does not use AASC to manufacture the QFT facts. Conventional field theory establishes the actions, currents, anomalies, topology, operator classes, tensor strata, and charge signatures first; the structural layer then controls target identity, lawful redescription, faithful quotienting, and the prohibition against promoting a derived survivor into a primitive symmetry principle. [1, 8]

The scientific proof spine is

fixed carrier and target family $\implies$ renormalizable accidental actions,
 accidental actions + SYM-3 anomaly/topology data $\implies$ quantum-sector-qualified charge directions,
 qualified directions + neutrino and operator support $\implies$ violation vectors,
 violation vectors $\implies$ Smith quotient + faithful generator map,
 target-local results + typed higher summaries $\implies$ type-indexed canonical atlas,
 type-indexed canonical atlas $\implies$ finite typed-result completeness ($31 + 1 + 1 + 1 = 34$).

No step is allowed to rewrite an upstream target retrospectively: adding ν_R , turning on M_R , admitting the Weinberg tensor, changing coefficient support, or entering a nontrivial electroweak sector creates a successor target.

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Part I

Scope, Dependencies, and Fixed Targets

1 Dependency order and position in the Symmetry Arc

The word *symmetry* carries several physically distinct burdens in the Standard Model. A transformation may act faithfully on the local algebra without being generated by a conserved quantum current; a classical current may acquire an anomaly; a perturbative selection rule may hold in one topological sector while transitions between sectors violate the associated charge; an operator basis may preserve a continuous direction at one dimension and break it at another; and a finite quotient may exist algebraically without yet defining an exact discrete gauge symmetry. SYM-7 is designed to keep those statements typed while following baryon and lepton charge through renormalizable, topological, neutrino, and effective-operator targets.

1.1 Upstream interfaces

The paper uses the earlier Symmetry Arc as a genuine dependency chain. The relevant interfaces are summarized in Table 1. Their function is not decorative citation: each fixes a piece of the comparison class that would otherwise be silently changed inside the present argument.

Two upstream statements are especially constraining. First, SYM-3 requires the downstream accidental-charge record to carry the lowest violating operator dimension, perturbative/nonperturbative

Table 1: Upstream Symmetry Arc interfaces used by SYM-7. The table records the scientific content imported into the present fixed-target classification.

Arc	Upstream result	Use in SYM-7	Boundary retained here
SYM-0	Symmetry-status calculus and target-preserving redescription [1]	Distinguishes physical action, current/diagnostic, obstruction, target update, and representative change	Does not supply sector-specific QFT data
SYM-1	Faithful gauge action and quotient discipline [2]	Fixes the physical algebra, gauge-center overlaps, and the rule that only faithful field action determines a quotient	Global-form completion remains open
SYM-2	Spacetime symmetry as domain anchor [3]	Keeps local operators, currents, and topological amplitudes in one fixed Poincaré/CPT domain	CPT is not reclassified here
SYM-3	Anomaly and electroweak-topology interface [4]	Supplies the $B + L$ current anomaly, integrated transfer law, and the separation of current anomaly from operator-order breaking	No thermal rate or cosmological history is imported
SYM-4	Discrete-action grammar [5]	Requires operator support, anomaly compatibility, and faithful action before a finite remnant is interpreted	No familiar discrete label is accepted as proof of exactness
SYM-5	Flavor and neutrino target bifurcation [6]	Fixes the Y_e , Y_ν , M_R , and κ strata that control family and total lepton symmetries	No cross-target flavor transport is assumed
SYM-6	Baryon/chiral normalization interface [7]	Fixes the faithful $U(1)_B$ normalization on the graded local algebra and keeps axial/chiral statements distinct from baryon charge	No electroweak conservation conclusion is imported from the QCD sector

status, neutrino target, anomaly class, and preserved remnant; operator-order effects may not be attributed retrospectively to the electroweak anomaly. Second, SYM-4 requires a discrete baryon/lepton remnant to import both anomaly status and discrete-action grammar: the breaking of a continuous phase is not by itself an exact discrete-symmetry theorem. Those constraints determine the architecture of Sections 4–7. [4, 5]

1.2 Division of evidentiary labor

The logical and evidentiary order are deliberately separated. Kernel necessity is logically prior to any determinate classification: a statement that two symmetry realizations are the same or different already presupposes fixed reference, admissible comparison, and consequence-bearing use. But the physical premises used in this paper are established by conventional QFT before the final structural disposition is reported. In particular,

- the Standard Model carrier and renormalizable grammar determine which field monomials exist;
- anomaly theory determines current nonconservation and topological selection rules;
- flavor tensors determine stabilizers in each neutrino stratum;
- published SMEFT bases determine the operator sectors used at dimensions five, six, seven, and nine;

- exact integer algebra determines the null lattices, Smith factors, and back-mapped remnant generators.

AASC then governs whether these facts belong to the same target, whether two presentations are lawful redescrptions, whether a quotient is faithful, and whether the resulting object has standing as an action, obstruction, diagnostic, transport, or derived remnant. This preserves the arc-wide rule that the structural classifier cannot create the physical antecedent it later classifies. [1, 8]

1.3 The exact SYM-7 burden

The present paper has four simultaneous obligations.

1. **Origin:** derive the accidental baryon and lepton actions from the admitted renormalizable grammar rather than assume them.
2. **Quantum qualification:** distinguish action, current anomaly, perturbative selection, and topological transfer without collapsing them into one “conserved/not conserved” bit.
3. **Operator-order transport:** identify the first registered baryon/lepton violating sectors and preserve their target, neutrino, and support qualifications.
4. **Remnant closure:** construct each continuous or finite survivor from the actual violation lattice, map it back to the physical fields, remove pre-existing overlaps, and stop at the global frontier when the data required for a stronger exactness claim are absent.

The final finite typed-result theorem is therefore not a statement that there are only 34 conceivable baryon/lepton symmetry questions. It is a completeness theorem for a 34-entry typed scientific ledger relative to the 13 registered target schemas (including the effect-set-indexed remnant schema), the target instances generated from the 12 admitted effect sets, and the specified complete or selected operator sectors. The ledger contains 31 target-local classes, one cross-target constraint, one target-family synthesis, and one integrated closure result; whole-arc candidate exhaustion remains the separate burden of SYM-11.

No backward target rewrite

A successor target may refine or extend an earlier one, but it does not retroactively alter the earlier theorem. The minimal massless-neutrino result remains a theorem of that target after a Dirac, Majorana, Weinberg, or higher-dimensional successor is studied. Likewise, a nontrivial electroweak topological sector does not rewrite the perturbative $Q_{EW} = 0$ statement; it supplies a distinct quantum-sector record.

2 Fixed targets, typed result objects, and faithful charge lattices

Accidental symmetries are often described as if they were fixed properties of a Lagrangian. That language is adequate only after the field content, operator grammar, flavor tensors, coefficient support, quantum sector, and faithful global action have been specified. Changing any of those ingredients can change the symmetry statement without contradiction. Baryon number is accidental in the renormalizable Standard Model but violated by higher-dimensional operators; family lepton numbers

occur in the massless-neutrino limit but are generically broken by flavor mixing; $B + L$ is respected by ordinary perturbative vertices in the trivial electroweak sector while its current is anomalous and topological transitions carry nonzero $\Delta(B + L)$.

2.1 Reference-complete targets

Let a target specification be

$$T = (\Sigma_{\text{SM}}, \Gamma, F_\nu, \mathcal{O}, Y, Q_{\text{EW}}, B_{\text{bg}}, \mathcal{E}_{\text{EFT}}, \lambda). \quad (1)$$

Here Σ_{SM} fixes the field content, Γ records the faithful gauge/global-form choice or an explicitly unresolved global frontier, F_ν chooses the neutrino sector, $\mathcal{O}$ and $\mathcal{E}_{\text{EFT}}$ fix operator support and order, Y and λ fix flavor-tensor data and coefficient stratum, Q_{EW} fixes the topological sector, and B_{bg} fixes admitted backgrounds. A change in any of these physical coordinates defines a different target rather than a new representative of the same one.

Table 2: Registered target schemas used in the classification. Each row fixes a combination of field content, neutrino structure, quantum/background sector, and operator support; the final row is parameterized by the admitted effect set S .

Target schema	Neutrino sector	Quantum/background sector	Active operator support	Role in the classification
Minimal renormalizable SM	no ν_R	$Q_{EW} = 0$	complete renormalizable SM grammar	classical accidental origin
Minimal SM with EW topology	no ν_R	integer electroweak sectors	renormalizable SM + topological amplitude	anomaly/topology transfer
Dirac ν_R extension	three ν_R , $M_R = 0$	trivial sector	renormalizable ν_R theory	Dirac-neutrino stabilizer
Dirac ν_R with EW topology	three ν_R , $M_R = 0$	integer electroweak sectors	Dirac theory + topological amplitude	electroweak specialization
Renormalizable Majorana ν_R	three ν_R , $M_R \neq 0$	fixed Majorana stratum	renormalizable Majorana term	$\Delta L = 2$ breaking
Weinberg EFT	minimal field content	fixed Weinberg stratum	nonzero $d = 5$ Weinberg operator	dimension-five LNV
Dimension-six SMEFT BNV	minimal field content	local EFT sector	four minimal $d = 6$ BNV types	minimal dimension-six BNV
Dimension-six ν SMEFT BNV	ν_R extension	local EFT sector	six BNV types	right-handed-neutrino extension
Dimension-seven SMEFT	minimal field content	local EFT sector	twelve LNV + six BLNV types	dimension-seven split
Selected dimension-nine sector	minimal field content	local EFT sector	complete $\Delta B = 2$ sector used here	baryon-number-two violation
Minimal $B - L$ background	minimal field content	$B - L$ and gravitational backgrounds	renormalizable SM grammar	local $B - L$ anomaly accounting
ν_R $B - L$ background	three ν_R	$B - L$ and gravitational backgrounds	$B - L$ -covariant ν_R support	local polynomial cancellation
Effect-set-indexed remnant schema $T(S)$	inherited from effect set S	effect-dependent	one of the twelve fixed effect sets	target instances for continuous and finite remnants

The first twelve rows specify fixed target schemas with the displayed parameter strata. The final row is a parameterized schema: supplying one of the twelve admitted effect sets produces a target instance $T(S)$. Accordingly, the phrase “13 target schemas” does not identify all remnant instances with one physical target.

The 13 rows of Table 2 are registered target schemas rather than thirteen unrelated theories. Twelve fix explicit carrier, tensor, background, topology, and operator-support strata; the final schema $T(S)$ is instantiated only after an admitted effect set S is supplied. The schemas are connected by explicit successor operations: add right-handed neutrinos, change the neutrino mass stratum, admit a nontrivial electroweak sector, add a specified effective-operator sector, switch on anomaly-testing backgrounds, or pass to an effect-set remnant instance. This is the same target/update discipline used throughout the earlier arc. [1, 4–6]

2.2 Typed result universe

The scientific outputs of SYM-7 are not all objects of one mathematical kind. The target-local actions, anomaly statuses, operator sectors, and individual remnant calculations possess one definite target and admit target-preserving comparison. A cross-target law, a synthesis over an indexed family of targets, and a whole-registry closure statement instead retain their declared scope and component data. Define the disjoint typed result universe

$$\mathfrak{R}_7 = \mathfrak{R}_{\text{local}} \sqcup \mathfrak{R}_{\text{constraint}} \sqcup \mathfrak{R}_{\text{family}} \sqcup \mathfrak{R}_{\text{closure}}. \quad (2)$$

The finite atlas in Part IV contains 31 target-local results, one cross-target constraint, one target-family result, and one integrated closure result. This typing preserves the distinction already built into the Symmetry Arc record calculus: a theorem whose domain is a family of targets is not forced into the same object category as a target-local physical action.

Definition 2.1 (Target-local equivalence). For $x, x' \in \mathfrak{R}_{\text{local}}$, write $x \sim_T x'$ when they have the same target, the same faithful action on the declared physical algebra, and identical perturbative, anomaly, topology, neutrino, support, remnant, and observable invariants, while differing only by an admissible weak-basis, charge-basis, operator-basis, current-improvement, Smith-presentation, or renormalization-scheme transformation. Charge and lattice transformations are required to be unimodular.

Definition 2.2 (Target-local canonical projection). For $x \in \mathfrak{R}_{\text{local}}$, define

$$\pi_T(x) = (T, X_{\text{faithful}}, F_{\text{support}}, P, A, Q, N, D, V, R, O), \quad (3)$$

where the entries encode the target, faithful charge action, operator support, perturbative status, anomaly data, topological sector, neutrino structure, dimension/charge signature, violation lattice, remnant, and observable class. The faithful-action component contains the acting lattice, exact generator or generator matrix, faithful image, physical algebra, and any kernel/quotient qualification. The projection is unchanged by admissible representative transformations and changes whenever a target-local class-defining physical invariant changes.

Definition 2.3 (Higher typed projections). For a cross-target constraint $c \in \mathfrak{R}_{\text{constraint}}$, set

$$\pi_{\text{constraint}}(c) = (\mathcal{D}_c, \varphi_c, \mathcal{C}_c, \mathcal{E}_c), \quad (4)$$

where $\mathcal{D}_c$ is the declared target family, φ_c the cross-target predicate, $\mathcal{C}_c$ its conclusion, and $\mathcal{E}_c$ its source or proof certificate. For a target-family result $f \in \mathfrak{R}_{\text{family}}$, set

$$\pi_{\text{family}}(f) = (I_f, (\pi_T(x_i))_{i \in I_f}, \Phi_f), \quad (5)$$

where I_f is the ordered component index and Φ_f the family-level conclusion. For an integrated closure result $z \in \mathfrak{R}_{\text{closure}}$, set

$$\pi_{\text{closure}}(z) = (I_z, (\pi_T(x_i))_{i \in I_z}, \mathcal{C}_z, \mathcal{N}_z), \quad (6)$$

where I_z records the component target instances or target-local results, $\mathcal{C}_z$ the closure conclusion, and $\mathcal{N}_z$ its exact nonclaims and successor frontier.

Definition 2.4 (Type-indexed canonical projection). Let $\tau(x) \in \{\text{local, constraint, family, closure}\}$ denote the result type. Define

$$\Pi_7(x) = \begin{cases} (\text{local}, \pi_T(x)), & x \in \mathfrak{R}_{\text{local}}, \\ (\text{constraint}, \pi_{\text{constraint}}(x)), & x \in \mathfrak{R}_{\text{constraint}}, \\ (\text{family}, \pi_{\text{family}}(x)), & x \in \mathfrak{R}_{\text{family}}, \\ (\text{closure}, \pi_{\text{closure}}(x)), & x \in \mathfrak{R}_{\text{closure}}. \end{cases} \quad (7)$$

Two typed results are canonically identical, written $x \equiv_7 x'$, exactly when $\Pi_7(x) = \Pi_7(x')$. Only the target-local sector is quotiented by $\sim_T$; the higher result types retain their declared scope and component projections.

The maps π_T and Π_7 are the reader-facing forms of the finite classification. The theorem in Part IV proves 31 distinct target-local $\sim_T$ classes and pairwise typed uniqueness for all 34 atlas entries.

2.3 Charge lattices and faithful quotients

The primitive lattice used for integer computations is

$$\Lambda_{\text{field}} = \mathbb{Z}\langle 3B, L_e, L_\mu, L_\tau \rangle, \quad \Lambda_{\text{phys}} = \mathbb{Z}\langle B, L_e, L_\mu, L_\tau \rangle. \quad (8)$$

The baryon normalization $N_B : \Lambda_{\text{field}} \rightarrow \Lambda_{\text{phys}}$ sends $3B \mapsto B$ by $\text{diag}(1/3, 1, 1, 1)$ while retaining its integral inverse on the image. This distinction matters whenever a finite Smith factor contains a baryonic direction: a quark carries physical baryon charge $1/3$, so the apparent finite action must be compared with the $SU(3)_c$ center on the elementary fields. Fermion parity is treated similarly. A finite factor is quotiented only when its full field action coincides with an already identified center or parity element, possibly after a continuous compensator; its order alone is not enough. This is the faithful-action discipline imported from SYM-1 and the discrete-action discipline imported from SYM-4. [2, 5]

Theorem 2.1 (Normalization and integral-presentation covariance). *The fixed baryon-normalization map $N_B : \Lambda_{\text{field}} \rightarrow \Lambda_{\text{phys}}$ relates two differently normalized charge lattices; it is not a unimodular basis change on one embedded integral lattice. Once either lattice is fixed, every admissible integral presentation change is implemented by an element of $GL(4, \mathbb{Z})$. Such changes preserve the violation row lattice up to lattice isomorphism, the quotient module, rank, integer null lattice, Smith invariant*

Table 3: Charge lattices, normalization, and faithful overlaps used in the remnant analysis.

Object	Definition	Role	Faithful identification
Λ_{field}	$\mathbb{Z}\langle 3B, L_e, L_\mu, L_\tau \rangle$	primitive integer lattice on elementary fields	none
Λ_{phys}	$\mathbb{Z}\langle B, L_e, L_\mu, L_\tau \rangle$	physical charge normalization on gauge-invariant states	color-singlet baryons have $B = 1$
N_B	$\text{diag}(1/3, 1, 1, 1)$	maps $3B \mapsto B$ while retaining the integral inverse on its image	baryonic phases are tested against the $SU(3)_c$ center quotient only after equality of the full field action is established
$Z(SU(3)_c)$ and $(-1)^F$	elementary-field actions of orders 3 and 2	remove finite factors that act only as an already identified gauge-center or fermion-parity element	

factors, and faithful elementary-field action; generator coordinates transform covariantly and are mapped through N_B only when physical baryon normalization is required.

Proof. In the displayed bases, $N_B = \text{diag}(1/3, 1, 1, 1)$ sends the primitive generator $3B$ to the physical generator B . Its determinant is $1/3$, so it is a fixed normalization between the two lattice embeddings rather than an element of $GL(4, \mathbb{Z})$ acting within either one. By contrast, an alternative integral presentation of a fixed lattice is related by a unimodular matrix M . Transport by M is an isomorphism of the full lattice: it carries the row lattice and its integer kernel bijectively, induces an isomorphism of the quotient module, and changes a violation matrix only by unimodular row or column operations. Rank and Smith invariant factors are therefore unchanged. Back-mapped generators are then evaluated on the elementary fields, so a presentation change cannot alter the faithful physical action. Preserving one selected charge vector without preserving this full lattice isomorphism would not be an admissible redescription. $\square$

2.4 Neutrino target graph

The lepton sector requires explicit target branching because the symmetry statement changes with the admitted tensors.

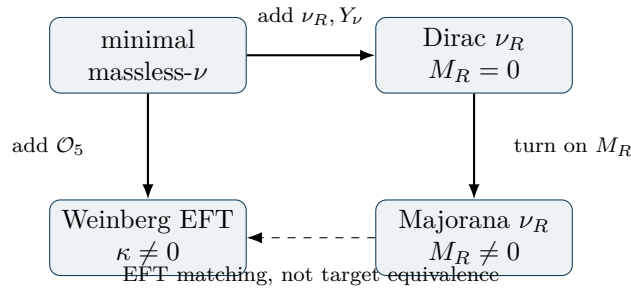


Figure 1: Neutrino target graph. Each arrow changes field or tensor support and therefore changes the target. The dashed arrow denotes an effective matching relation, not equivalence of the underlying targets.

2.5 Scientific construction order

The classification proceeds in the order shown in Figure 2. The sequence is substantive: remnants are computed only after the violating effects are fixed, and finite factors are interpreted physically only after the corresponding field action and pre-existing overlaps have been identified.

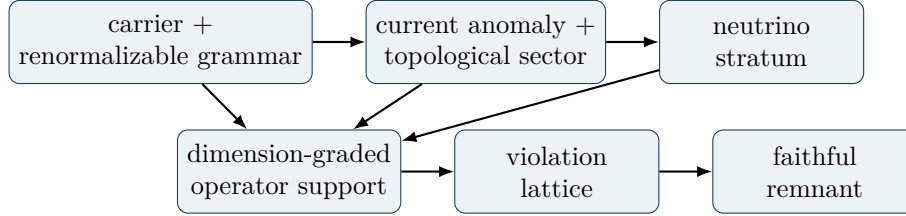


Figure 2: Scientific construction order. Target data determine the admissible symmetry actions and effect support; violation lattices are constructed from those effects, and physical remnants are identified only after faithful quotienting and anomaly qualification.

Primitive-data prohibition

No final symmetry label, remnant name, or desired class count is an input to the physical construction. The inputs are carrier data, tensor strata, anomaly/topology facts, published operator sectors, and exact charge actions. The labels “baryon number”, “ $B - L$ ”, or a familiar discrete name receive physical standing only through the action and invariants they denote on the fixed target.

Part II

Physical Accidental-Charge Construction

3 The renormalizable origin of accidental charge

3.1 Monomial exhaustion

For the minimal field target, the relevant classical grammar consists of gauge, fermion, and Higgs kinetic terms, the Higgs potential, and the three Yukawa classes. In left-handed Weyl notation the charges of $Q, u^c, d^c, L_i, e_i^c, H$ make each declared monomial neutral under B and, before neutrino masses, under the independent family phases. Gauge invariance and canonical dimension exclude a renormalizable color-singlet monomial with nonzero baryon charge. The conclusion is therefore a consequence of the admitted grammar; it is not a primitive law.

The ν_R carrier extension changes the grammar. Its neutral ν^c kinetic term is a separate target-dependent class, so the extended theory does not reuse the seven-row minimal grammar by omission. A Dirac Yukawa tensor Y_ν preserves a diagonal total lepton phase but generically removes family differences by misalignment with Y_e . The gauge-singlet Majorana monomial $\nu^c \nu^c$ has dimension three and carries $\Delta L = 2$. Hence total L in a ν_R theory requires the explicit stratum $M_R = 0$; it cannot be inferred merely from the presence of right-handed neutrinos.

Theorem 3.1 (Minimal renormalizable monomial exhaustion). *The seven minimal grammar classes*

Table 4: Renormalizable monomial classes relevant to the accidental-charge analysis. Every listed class is neutral under baryon number; in the minimal massless-neutrino target the charged-lepton terms also preserve the three family phases.

Class	Schematic fields	Accidental charge	Target condition
Gauge kinetic	$B_{\mu\nu}, W_{\mu\nu}, G_{\mu\nu}$	0	all targets
Fermion kinetic	Q, u^c, d^c, L, e^c	0	all minimal-carrier targets
Singlet-neutrino kinetic	ν^c	0	targets containing ν_R
Higgs kinetic	$H, H^\dagger$	0	all targets
Higgs potential	$H, H^\dagger$	0	all targets
Up-type Yukawa	$QH u^c$	0	generic Y_u
Down-type Yukawa	$QH^\dagger d^c$	0	generic Y_d
Charged-lepton Yukawa	$LH^\dagger e^c$	0	generic Y_e

in Table 4 exhaust the gauge-invariant canonical-dimension- ≤ 4 monomials relevant to the accidental-charge span considered here.

Proof. Kinetic and potential terms occur in conjugate-neutral combinations. A Lorentz scalar Yukawa monomial contains two Weyl fermions and one Higgs, and hypercharge and nonabelian singlet conditions select the u , d , and charged-lepton Yukawa classes. Purely fermionic Lorentz scalars require at least two fermions; a baryon-violating color singlet requires at least three quarks and then a fourth fermion for Lorentz invariance, giving dimension six. The enumerated minimal grammar is therefore complete at $d \leq 4$ within the target. $\square$

Theorem 3.2 (Minimal baryon accidental-origin theorem). *The complete minimal renormalizable Lagrangian is invariant under the faithful baryon action $U(1)_B$ on the declared local algebra.*

Proof. Each grammar row has zero $3B$ charge. The quark action is normalized to physical $B = 1/3$ only after the color-center identification of Table 3. The monomial-exhaustion result excludes any omitted same-target renormalizable monomial carrying baryon charge. This proves classical accidental invariance, not quantum current conservation across electroweak topology. $\square$

Theorem 3.3 (Minimal lepton-family accidental-origin theorem). *In the generic minimal massless-neutrino target, the renormalizable action has three independent family lepton phases.*

Proof. With only Y_e , a weak-basis transformation diagonalizes the charged-lepton tensor, and the stabilizer contains independent simultaneous phases of each L_i and e_i^c . There is no Y_ν , M_R , or κ in this target. The family phases are therefore neutral directions of the complete grammar. Their topology status is treated separately in Section 4. $\square$

Theorem 3.4 (Right-handed-neutrino restriction theorem). *For three ν_R , generic Dirac Yukawa support with $M_R = 0$ preserves total L and generically breaks family differences; unrestricted renormalizable support admits $M_R \neq 0$ and breaks total L by two.*

Proof. Assign $L(\nu_R) = 1$, equivalently $L(\nu^c) = -1$. The target-guarded ν^c kinetic term is neutral, $LH\nu^c$ preserves the diagonal phase, and $\nu^c\nu^c$ has charge -2 . Generic misaligned Y_e and Y_ν have only

their common diagonal stabilizer. Thus the Dirac conclusion is stratum-qualified, the Majorana guard is a distinct target, and the right-handed-neutrino specialization explicitly adds three singlet neutrinos rather than silently changing the field content. $\square$

Fixed-target test of the origin theorem

A single missing gauge-invariant canonical-dimension- ≤ 4 monomial carrying nonzero B or claimed L_i charge in the same target would falsify the preceding minimal-target results. Adding ν_R , M_R , or the Weinberg tensor is not such a falsifier; each changes the target fingerprint.

4 Current anomalies, quantum sectors, and electroweak topology

4.1 Four statements that must not collapse

For a classically defined action one may construct a Noether current, ask whether a local counterterm representative has an anomalous divergence, ask whether ordinary perturbation theory in the $Q_{\text{EW}} = 0$ sector obeys a selection rule, and ask how amplitudes between different topological sectors transform. These are four different predicates. The electroweak anomaly relates them, but does not identify them.

With $N_g = 3$ and integer electroweak charge Q_{EW} , the imported zero-mode theorem gives

$$\Delta B = N_g Q_{\text{EW}}, \quad \Delta L_i = Q_{\text{EW}}, \quad \Delta L = N_g Q_{\text{EW}}. \quad (9)$$

Thus the physical transfer row is $v_{\text{EW}} = (3, 1, 1, 1)$ in (B, L_e, L_μ, L_τ) units, and its integral field-lattice lift is $(9, 1, 1, 1)$. The equation is an integrated selection rule. It neither converts the effect into an ordinary SMEFT Wilson coefficient nor determines a thermal transition rate [9–11].

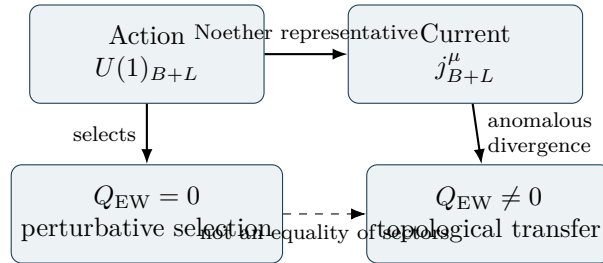


Figure 3: Four-level $B + L$ diagram. The action is physical, its current is anomalous, ordinary local perturbative vertices in the trivial sector respect the selection rule, and nontrivial topology transfers $B + L$.

The anomaly calculation may be written directly as a pairing between each physical current vector in (B, L_e, L_μ, L_τ) and the appropriate anomaly functional. For the electroweak selection rule, pairing with v_{EW} is nonzero for B, L_e, L_μ, L_τ separately and vanishes for $B - L$. This formulation makes the null directions consequences of the physical charge vectors rather than conventions attached to current names.

4.2 Null lattice and topological vertex

The null lattice is the integer kernel of v_{EW} . A convenient physical basis is

$$n_1 = (1, -1, -1, -1), \quad n_2 = (0, 1, -1, 0), \quad n_3 = (0, 0, 1, -1). \quad (10)$$

Table 5: Status of the principal baryon- and lepton-number directions across the renormalizable, electroweak-topological, and higher-operator layers.

Direction	Renormalizable origin	Electroweak current/topology	Additional operators	Classification
B	Accidental in the minimal $d \leq 4$ theory	Current anomalous; $\Delta B = 3Q_{\text{EW}}$	Violated by BNV sectors	Derived, target-dependent
L	Accidental when the relevant neutrino tensors are absent	Current anomalous; $\Delta L = 3Q_{\text{EW}}$	Broken by Majorana and LNV support	Neutrino- and support-dependent
$B + L$	Accidental where total L is defined	Electroweak transfer $\Delta(B + L) = 6Q_{\text{EW}}$	Further target-dependent breaking	Not exact across nontrivial topology
$B - L$	Target-dependent	Electroweak-null	Preserved or broken according to operator support	Continuous only where all active effects preserve it
$L_e - L_\mu$, $L_\mu - L_\tau$	Present in the massless-neutrino limit	Electroweak-null	Generically broken by flavor support	Flavor-stratum dependent

Table 6: Anomaly statements used in the classification. Electroweak current nonconservation and local $B - L$ background anomalies are kept distinct.

Current / action	Probe	Target	Result
$B - L$	$SU(3)_c$, $SU(2)_L$, $U(1)_Y$ mixed local anomalies	minimal SM and ν_R extensions	mixed dynamical-gauge coefficients vanish
$B + L$	electroweak $SU(2)_L$ topology	minimal and Dirac targets	anomalous current; $\Delta(B + L) = 6Q_{\text{EW}}$
$B - L$	mixed gravitational background	minimal SM	-1 per generation in the stated left-handed convention
$B - L$	cubic background	minimal SM	-1 per generation in the stated left-handed convention
$B - L$	gravitational + cubic backgrounds	one ν_R per family	local polynomial deficits cancel

Table 7: Electroweak transfer vectors and a primitive basis of the integer null lattice.

Object	Vector	Basis	Interpretation
Physical transfer	$(3, 1, 1, 1)$	(B, L_e, L_μ, L_τ)	$\Delta B = 3Q_{\text{EW}}$ and $\Delta L_i = Q_{\text{EW}}$
Primitive transfer	$(9, 1, 1, 1)$	$(3B, L_e, L_\mu, L_\tau)$	integral lift to Λ_{field}
Null generator	$(1, -1, -1, -1)$	physical lattice	$B - L$
Null generator	$(0, 1, -1, 0)$	physical lattice	$L_e - L_\mu$
Null generator	$(0, 0, 1, -1)$	physical lattice	$L_\mu - L_\tau$

These generate $B - L$, $L_e - L_\mu$, and $L_\mu - L_\tau$. Direct dot products with v_{EW} vanish. The first statement is independent of neutrino flavor tensors; the family-difference actions themselves exist generically only in the massless-neutrino target.

For $Q_{EW} = 1$, the zero modes produce nine quark-doublet and three lepton-doublet insertions. Twelve Weyl fields have canonical dimension $12 \times 3/2 = 18$. This is the dimension-18 electroweak 't Hooft vertex; its effect type is topological, not an ordinary SMEFT Wilson operator. The common dimension is not a license to identify the two categories.

Theorem 4.1 (Current/action/conservation separation). *An anomalous divergence of a current does not erase the corresponding action on the local algebra, and it does not imply an ordinary perturbative violating vertex.*

Proof. The action is defined by its faithful automorphism of fields and local operators. The current representative and its divergence depend additionally on the quantum measure and the chosen counterterm class. The trivial-sector perturbative selection rule is a statement about matrix elements inside $Q_{EW} = 0$, whereas the integrated anomaly compares sectors. These objects occupy separate fields X, A, P, Q and separate status coordinates, so no inference can silently cross the distinction. $\square$

Theorem 4.2 (Electroweak charge-transfer theorem). *For three generations, a unit electroweak topological transition has transfer vector $(3, 1, 1, 1)$ in the physical charge basis and $(9, 1, 1, 1)$ in the primitive field basis.*

Proof. The electroweak zero-mode count supplies one lepton-doublet mode per generation and three colored quark-doublet modes per generation. Physical baryon normalization assigns $1/3$ to each quark, giving $\Delta B = 3$, while each family receives $\Delta L_i = 1$. Applying the inverse baryon normalization lifts the first coordinate to nine. The two vectors are therefore exact images of one another under baryon normalization. $\square$

Theorem 4.3 ($B + L$ anomaly theorem). *The minimal and Dirac targets possess a physical $B + L$ action whose current is electroweak anomalous and whose topological transfer is $\Delta(B + L) = 6Q_{EW}$, while ordinary local SM vertices in the trivial sector remain selecting.*

Proof. Sum the first four entries of Equation (9): $\Delta(B + L) = 3Q_{EW} + 3Q_{EW}$. The renormalizable and neutrino-sector results supply the classical actions in their respective targets, while the current/action distinction prevents the anomalous divergence from being reinterpreted as a local perturbative vertex. The conclusion is target- and sector-indexed. $\square$

Theorem 4.4 (Electroweak nullspace theorem). *The integer null lattice of electroweak transfer has rank three and is generated by $B - L$, $L_e - L_\mu$, and $L_\mu - L_\tau$ in the physical basis used here.*

Proof. The nonzero 1×4 row v_{EW} has rank one over $\mathbb{Z}$ because its entries have gcd one. Its kernel is therefore free of rank three. The displayed n_i are integral, independently span a primitive rank-three sublattice, and satisfy $v_{EW} n_i^T = 0$. The 3×3 minor on the (B, L_e, L_μ) columns has determinant one, proving that the generated rank-three sublattice is primitive and hence equals the full integer kernel. $\square$

Theorem 4.5 (Topological-vertex noncollapse theorem). *The dimension-18 electroweak 't Hooft vertex is a topological amplitude class and cannot be substituted for a perturbative SMEFT Wilson operator.*

Proof. The vertex is fixed by zero modes and topological charge, not by choosing an independent local Wilson coefficient in a SMEFT basis. It is supported only as a nonperturbative topological amplitude in a target with nontrivial Q_{EW} . Replacing it by an ordinary SMEFT Wilson operator would therefore change both the effect type and the physical sector, not merely the notation. $\square$

Summary. Electroweak topology violates $B+L$ across sectors, preserves the $B-L$ transfer direction, and leaves two family-difference null vectors at the selection-matrix level. Nothing in this section computes a sphaleron rate or a baryogenesis history.

5 Neutrino fibers and the qualified $B-L$ boundary

5.1 Four inequivalent neutrino targets

The four cases in Figure 1 answer different questions. The minimal massless-neutrino target has no dynamical ν_R and no Weinberg tensor. The Dirac target has three ν_R , a generic Y_ν , and the explicit stratum $M_R = 0$. The renormalizable Majorana target turns on a generic symmetric M_R at dimension three. The Weinberg EFT instead keeps the minimal field content but turns on a generic symmetric tensor κ at dimension five. Matching a heavy Majorana theory onto the Weinberg target can be physically useful, but it is not representative equivalence because carrier, operator order, and evidence change.

Table 8: The four neutrino targets and their generic continuous lepton symmetries. Total L in the minimal target is the diagonal combination of the three family phases, not an additional fourth generator.

Fiber	Carrier	Flavor tensors	M_R	κ	Generic lepton symmetry
minimal	minimal SM	Y_e	absent	absent	three family phases $U(1)_{L_e} \times U(1)_{L_\mu} \times U(1)_{L_\tau}$; total $\bar{L}$ is the diagonal combination
Dirac ν_R	SM + $3\nu_R$	Y_e, Y_ν	0	absent	total L ; family differences generically broken by misalignment
Majorana ν_R	SM + $3\nu_R$	Y_e, Y_ν, M_R	generic nonzero	absent	continuous total L broken by $\Delta L = 2$
Weinberg EFT	minimal SM carrier	Y_e, κ	absent	generic nonzero	continuous total L and generic family differences broken

Theorem 5.1 (Four-fiber neutrino separation). *The minimal, Dirac, renormalizable-Majorana, and Weinberg-EFT cases are pairwise target-inequivalent and exhaust the neutrino branches used in this paper.*

Proof. Their carrier fields, Y_ν support, M_R support, κ support, and EFT order differ in at least one fingerprint coordinate. No licensed representative transport changes any of those target coordinates. None of the admissible representative transformations changes these carrier or operator-support coordinates, so the four cases remain distinct targets. $\square$

Theorem 5.2 (Dirac total-lepton theorem). *In the generic three- ν_R Dirac target with $M_R = 0$, total lepton number is continuous while nontrivial family differences are generically broken.*

Proof. The diagonal phase rotating L, e_R, ν_R together commutes with both Y_e and Y_ν . A family-difference generator must instead stabilize both tensors. Generic misalignment leaves only the common diagonal $U(1)$, so an enlarged family stabilizer requires a texture witness. Neutrino oscillation or Dirac mixing therefore does not by itself violate total L . $\square$

Theorem 5.3 (Renormalizable Majorana theorem). *A generic nonzero M_R is a dimension-three, $\Delta L = 2$ effect that removes continuous total lepton number in the renormalizable Majorana target.*

Proof. The singlet bilinear is gauge invariant and has canonical dimension three. Under the total lepton phase its two ν^c factors carry charge -2 , so a fixed nonzero M_R is not invariant. A spurionic transformation would be a different statement; the target fixes the tensor rather than promoting it to a dynamical compensator. $\square$

Theorem 5.4 (Weinberg target theorem). *A generic nonzero symmetric Weinberg tensor is a dimension-five, $\Delta L = 2$ effect that removes continuous total L and generic family differences in the Weinberg EFT without retroactively changing the minimal renormalizable result.*

Proof. The operator $(L^T C L) H H$ carries two units of lepton number and its generic flavor tensor has no continuous family stabilizer. It is absent from the renormalizable minimal target and present in the dimension-five successor. The minimal-target statement therefore remains distinct, while the unique-class literature result fixes the operator sector [12]. $\square$

5.2 Local polynomial $B - L$ accounting

All anomaly coefficients use left-handed Weyl fields and a fixed common index convention. In the minimal target, mixed anomalies of $B - L$ with the dynamical SM gauge factors cancel, but the local cubic and mixed-gravitational background sums per generation equal -1 in the chosen orientation. Three generations give -3 . Adding one right-handed neutrino with $L(\nu_R) = 1$ and hence $(B - L)(\nu_R) = -1$ per family—represented in the left-handed Weyl sum by ν^c with $(B - L)(\nu^c) = +1$ —cancels both deficits. This is a statement about a local anomaly polynomial. Faithful global form, scalar breaking, line/defect data, global anomalies, and the exact discrete gauge remnant require a successor target.

$$\mathcal{A}_{\text{grav}}(B - L)|_{\text{minimal}} = -N_f, \quad \mathcal{A}_{(B-L)^3}|_{\text{minimal}} = -N_f, \quad (11)$$

$$\mathcal{A}_{\text{grav}}(B - L)|_{N_f \nu_R} = 0, \quad \mathcal{A}_{(B-L)^3}|_{N_f \nu_R} = 0. \quad (12)$$

The signs depend on the declared left-handed orientation; cancellation and noncancellation do not.

Theorem 5.5 (Minimal $B - L$ local-gauge theorem). *In the minimal SM target, the mixed local anomaly coefficients of $B - L$ with the dynamical $SU(3)_c$, $SU(2)_L$, and $U(1)_Y$ gauge factors vanish in the stated left-handed convention.*

Proof. The one-generation left-handed Weyl sum is performed representation by representation with multiplicities. Quark and lepton contributions cancel in each dynamical-gauge column. The cancellation follows by the ordinary one-generation left-handed Weyl sum. The theorem says nothing yet about gravity or a background $B - L$ gauge field. $\square$

Theorem 5.6 (Minimal cubic/gravitational obstruction theorem). *The same minimal target has nonzero mixed-gravitational and cubic $B - L$ background coefficients, equal to -1 per generation in the stated orientation.*

Proof. Summing $B - L$ and $(B - L)^3$ over the minimal left-handed spectrum leaves exactly the missing singlet-neutrino contribution. The three-generation deficits are therefore -3 . Because the background probe is explicit in the target, this obstruction cannot be erased by quoting only the dynamical-gauge columns. $\square$

Theorem 5.7 (Three- ν_R local cancellation theorem). *With three right-handed neutrinos satisfying $L(\nu_R) = 1$ (equivalently, three left-handed conjugates with $(B - L)(\nu^c) = +1$ in the anomaly sum) and $B - L$ -covariant support, the local cubic and mixed-gravitational polynomial deficits cancel.*

Proof. Each added left-handed conjugate contributes $+1$ to both sums and cancels the corresponding per-family deficit. The condition is one singlet per family in this normalization, consistent with the sterile-count result [13]. A fixed generic M_R would explicitly break continuous $B - L$ and belongs to the renormalizable Majorana target instead. $\square$

Theorem 5.8 (Gauging/global frontier theorem). *Local polynomial cancellation is necessary but not sufficient to assert a gauged or globally complete $B - L$ symmetry; those assertions require additional global-form and breaking-sector data.*

Proof. Gaugeability depends on the faithful global group and charge lattice, breaking-sector fields, discrete quotient, line and defect content, and global anomalies. None of those data is fixed by the two local polynomial sums alone, so the present result stops at local anomaly cancellation. $\square$

Summary. Neutrino mixing, renormalizable Majorana breaking, and Weinberg-operator breaking belong to different targets. Minimal $B - L$ is electroweak-null and mixed-SM-gauge-null but has local cubic and gravitational deficits; three ν_R remove those polynomial deficits without settling the global completion problem.

Part III

Operator Dimension and Remnant Structure

6 Operator dimension and exact charge signatures

6.1 Signature schema and support

Each operator used in the classification carries

$$(\text{id}, T, d, \text{effect type}, \text{fields}, \text{gauge contraction}, \text{Lorentz class}, \text{flavor tensor}, \Delta B, \Delta L, \text{support}, \text{basis}, \text{scheme}, \text{source}). \quad (13)$$

Charge signatures are exact rational/integer data. Coefficient support is a target coordinate: generic nonzero support enters a violation lattice, exact zero defines a distinct stratum, unspecified support is not silently promoted to nonzero, and a nonperturbative topological amplitude is a separate effect type.

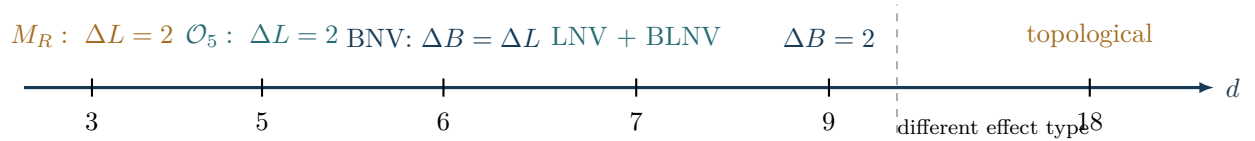


Figure 4: Operator-dimension ladder. Dimensions 3, 5, 6, 7, and selected 9 are local operator sectors; the dimension-18 electroweak vertex is topological and separated by effect type.

Table 9: Operator and topological sectors entering the charge classification. Counts are operator types before Hermitian conjugation unless stated otherwise.

d	Sector	$(\Delta B, \Delta L)$	Types	Role in the classification
3	Renormalizable Majorana mass	$(0, 2)$	1	Breaks continuous total lepton number when $M_R \neq 0$
5	Weinberg operator	$(0, 2)$	1	Unique dimension-five SMEFT lepton-number-violating class [12]
6	Minimal SMEFT BNV	$(1, 1)$	4	Warsaw-basis baryon-number violation; preserves $B - L$ [14]
6	ν_R -extended BNV	$(1, 1)$	2	Additional right-handed-neutrino operator types [15]
7	LNV component	$(0, 2)$	12	Complete lepton-number-violating component [16]
7	BLNV component	$(1, -1)$	6	Complete baryon-and-lepton-number-violating component [16]
9	Selected $\Delta B = 2$ sector	$(2, 0)$	5	Complete selected charge sector, excluding Hermitian conjugates [17]
18	Electroweak 't Hooft vertex	$(3, 3)$	1 class	Topological amplitude, not an ordinary SMEFT Wilson operator [9, 10]

6.2 Dimension five and six

The unique dimension-five SMEFT class can be written schematically as

$$\mathcal{O}_5^{pr} = \epsilon_{ij}\epsilon_{mn}(L_p^{iT} C L_r^m) H^j H^n, \quad \kappa_{pr} = \kappa_{rp}, \quad (14)$$

and has $(\Delta B, \Delta L) = (0, 2)$ [12]. At dimension six the minimal-field BNV sector contains four Warsaw types. With three right-handed neutrinos, two further types occur. Their published symbols, field multisets, and contraction classes are retained rather than replaced by an invented compressed basis [14, 15].

Table 10: Complete dimension-six baryon-number-violating operator types used here. The first four are the minimal Warsaw-basis classes; the final two require right-handed neutrinos.

Operator	Schematic fields	Gauge contraction	ΔB	ΔL	Carrier
Q_{duql}	$duq\ell$	$\epsilon_c \epsilon_2$	1	1	minimal / ν_R extension
Q_{qque}	$qque$	$\epsilon_c \epsilon_2$	1	1	minimal / ν_R extension
Q_{qqql}	$qqq\ell$	$\epsilon_c \epsilon_2 \epsilon_2$	1	1	minimal / ν_R extension
Q_{duue}	$duue$	ϵ_c	1	1	minimal / ν_R extension
Q_{qqdN}	$qqdN$	$\epsilon_c \epsilon_2$	1	1	ν_R extension only
Q_{uddN}	$uddN$	ϵ_c	1	1	ν_R extension only

The minimal four and the two additional ν_R types all have $(\Delta B, \Delta L) = (1, 1)$ in the convention used here and therefore preserve $B - L$. The ν_R types do not belong to the minimal carrier target. Hermitian conjugates reverse both signs without changing the preserved direction.

6.3 The corrected dimension-seven split

The complete dimension-seven sector relevant here has two charge components: twelve types with $(0, 2)$ and six types with $(1, -1)$, before Hermitian conjugation and with flavor tensors retained. The second component therefore preserves $B + L$, not $B - L$. This sign fixes the corresponding violation row and therefore the resulting remnant [16].

Table 11: Complete dimension-seven operator types in the two charge components used here: twelve LNV types with $(\Delta B, \Delta L) = (0, 2)$ and six BLNV types with $(1, -1)$, following Liao and Ma [16].

Operator	Schematic fields	Contraction class	ΔB	ΔL
$\mathcal{O}_{LH}$	$LLHHHH^\dagger$	two weak ϵ tensors and $H^\dagger H$ singlet	0	2
$\mathcal{O}_{LeHD}$	$LeHHHD$	weak ϵ contractions	0	2
$\mathcal{O}_{LHD1}$	$LLHHDD$	first independent weak-index contraction	0	2
$\mathcal{O}_{LHD2}$	$LLHHDD$	second independent weak-index contraction	0	2
$\mathcal{O}_{LHB}$	$LLHHB_{\mu\nu}$	weak singlet contraction	0	2
$\mathcal{O}_{LHW}$	$LLHHW_{\mu\nu}^I$	weak triplet contraction	0	2
$\mathcal{O}_{duLLD}$	$\bar{d}uLLD$	weak ϵ contraction	0	2
$\mathcal{O}_{eLLLH}$	$\bar{e}LLLH$	two weak ϵ contractions	0	2
$\mathcal{O}_{dLQLH1}$	$\bar{d}LQLH$	first independent weak-index contraction	0	2
$\mathcal{O}_{dLQLH2}$	$\bar{d}LQLH$	second independent weak-index contraction	0	2
$\mathcal{O}_{dLueH}$	$\bar{d}LueH$	weak ϵ contraction	0	2
$\mathcal{O}_{QuLLH}$	$\bar{Q}uLLH$	weak ϵ contraction	0	2
$\mathcal{O}_{Ldu\tilde{d}\tilde{H}}$	$\bar{L}du\tilde{d}\tilde{H}$	color ϵ and weak contraction	1	-1
$\mathcal{O}_{LdddH}$	$\bar{L}dddH$	color ϵ and weak singlet	1	-1
$\mathcal{O}_{eQdd\tilde{H}}$	$\bar{e}Qdd\tilde{H}$	color ϵ and weak singlet	1	-1
$\mathcal{O}_{LdQQ\tilde{H}}$	$\bar{L}dQQ\tilde{H}$	color ϵ and weak singlet	1	-1
$\mathcal{O}_{LQddD}$	$\bar{L}QddD$	color ϵ and weak singlet	1	-1
$\mathcal{O}_{edddD}$	$\bar{e}dddD$	color ϵ contraction	1	-1

6.4 Selected complete dimension-nine $\Delta B = 2$ sector

For the fixed sector $(\Delta B, \Delta L) = (2, 0)$, the dimension-nine classification contains five operator types without Hermitian conjugates and ten with them. The paper claims completeness only for that selected charge sector, not for the entire dimension-nine SMEFT basis [17].

Table 12: Complete selected dimension-nine $(\Delta B, \Delta L) = (2, 0)$ sector used in this paper, without Hermitian conjugates, following Liao and Ma [17].

Operator	Schematic fields	Independent contraction	ΔB	ΔL
$\mathcal{O}_{ddQQQQ}^{(1)}$	$d d Q Q Q Q$	first color-pair / weak-pair contraction	2	0
$\mathcal{O}_{ddQQQQ}^{(2)}$	$d d Q Q Q Q$	second color-pair / weak-pair contraction	2	0
$\mathcal{O}_{udddQQ}$	$u d d d Q Q$	double color ϵ with weak pair	2	0
$\mathcal{O}_{uddddd}^{(1)}$	$u u d d d d$	first double-color- ϵ contraction	2	0
$\mathcal{O}_{uddddd}^{(2)}$	$u u d d d d$	second double-color- ϵ contraction	2	0

6.5 Dimension/charge parity

For a local Lorentz- and SM-gauge-invariant operator built from the SM fields used here, the parity theorem of Kobach [18] states

$$d \equiv \frac{\Delta B - \Delta L}{2} \pmod{2}, \quad \Delta B - \Delta L \in 2\mathbb{Z}. \quad (15)$$

Equation (15) is evaluated for every local operator in Table 13. The electroweak topological vertex is shown for comparison but is excluded from this local-operator theorem because it belongs to a different effect type.

Table 13: Dimension/charge parity by local-operator sector. Each individual operator in the corresponding source-qualified sector satisfies the same parity relation.

Sector	d	ΔB	ΔL	$d - (\Delta B - \Delta L)/2 \pmod{2}$
Renormalizable Majorana mass	3	0	2	0
Weinberg operator	5	0	2	0
Minimal and ν_R BNV	6	1	1	0
Dimension-seven LNV	7	0	2	0
Dimension-seven BLNV	7	1	-1	0
Selected $d = 9, \Delta B = 2$	9	2	0	0

Theorem 6.1 (Operator-signature theorem). *Every admitted local operator has a total, target-compatible signature, and every violation row used later is derived from those signatures and the stated flavor support.*

Proof. The field content, flavor tensor, and coefficient stratum determine the charge signature and the family-resolved violation rows. Generic tensor support expands total ΔL into the family directions appropriate to that target. Zero or unspecified coefficient support is not included in a fixed nonzero effect set. $\square$

Theorem 6.2 (Dimension-five uniqueness theorem). *Within the minimal SMEFT target, the Weinberg class is the unique dimension-five operator type and carries $(\Delta B, \Delta L) = (0, 2)$.*

Proof. Uniqueness follows from the classification theorem of Weinberg [12]. The local specialization checks the minimal carrier set, dimension five, symmetric flavor tensor, and gauge/Lorentz contractions. Its exact charge follows by summing two lepton-doublet charges and two neutral Higgs charges. $\square$

Theorem 6.3 (Dimension-six BNV theorem). *The complete minimal-field dimension-six BNV sector consists of the four Warsaw types, each with $\Delta B = \Delta L$ and hence $\Delta(B - L) = 0$.*

Proof. The Warsaw basis contains the four types Q_{duql} , Q_{qqe} , $Q_{qq\ell}$, and Q_{duue} . Summing the field charges of each type gives $(\Delta B, \Delta L) = (1, 1)$. $\square$

Theorem 6.4 (ν_R dimension-six extension theorem). *The three- ν_R carrier extension adds exactly the two BNV types Q_{qqdN} and Q_{uddN} at dimension six, without changing the minimal four.*

Proof. The ν SMEFT classification supplies two additional types containing N [15]. Their carrier requirement is available only in the right-handed-neutrino EFT target; inserting either into the minimal field content is outside the operator basis being classified. Both carry the same $(1, 1)$ signature as the minimal BNV component. $\square$

Theorem 6.5 (Dimension-seven split theorem). *The dimension-seven basis splits into twelve $\Delta B = 0, \Delta L = 2$ types and six $\Delta B = 1, \Delta L = -1$ types.*

Proof. The primary-source classification gives disjoint lists of length 12 and 6. Every member has dimension seven and the corresponding charge signature. Hermitian conjugation reverses the charge row but generates the same violation sublattice; completeness is the cited field-theoretic classification theorem. $\square$

Theorem 6.6 (Dimension-nine $\Delta B = 2$ theorem). *The selected complete $d = 9$, $(\Delta B, \Delta L) = (2, 0)$ sector contains five types without Hermitian conjugates.*

Proof. The selected source table contains two $\mathcal{O}_{ddQQQQ}$ contractions, $\mathcal{O}_{udddQQ}$, and two $\mathcal{O}_{ududdd}$ contractions. Each has six quark fields, dimension nine, and charge $(2, 0)$. The completeness statement is restricted throughout to this selected charge sector. $\square$

Theorem 6.7 (Dimension-parity theorem). *All local operators in the stated scope obey Equation (15); the dimension-18 topological class is not used as a local-operator counterexample.*

Proof. Kobach's theorem supplies the analytic relation under the local SM-field hypotheses [18]. Exact arithmetic gives odd parity for M_R , $\mathcal{O}_5$, both dimension-seven components, and the selected dimension-nine sector, and even parity for both dimension-six sectors. Table 13 records the resulting parity test for every local operator in the stated scope. $\square$

Summary. Operator dimension, charge signature, field content, flavor tensor, coefficient support, and effect type are treated as one target-dependent object. The dimension ladder is finite and source-qualified only in the sectors stated above.

7 Violation lattices, Smith forms, and physical remnants

7.1 The mathematical and physical objects

For a fixed effect set S , let V_S be the integer matrix whose rows are the induced charge transfers in the primitive basis $(3B, L_e, L_\mu, L_\tau)$. Its row lattice is

$$L_V(S) = \text{span}_{\mathbb{Z}}\{\text{rows of } V_S\} \subset \Lambda_{\text{field}}. \quad (16)$$

The character group preserving all effects is

$$H_{\text{pres}}(S) = \text{Hom}(\Lambda_{\text{field}}/L_V(S), U(1)). \quad (17)$$

If the nonzero Smith factors are $s_1 \mid \cdots \mid s_r$, then

$$\Lambda_{\text{field}}/L_V(S) \cong \mathbb{Z}^{4-r} \oplus \bigoplus_{s_i > 1} \mathbb{Z}_{s_i}. \quad (18)$$

This is only the presentation group. Let $\mathcal{F}_T$ denote the elementary-field carrier of the target and define

$$G_S = \text{Im}[H_{\text{pres}}(S) \curvearrowright \mathcal{F}_T]. \quad (19)$$

Let $\mathcal{O}_T \subset \text{Aut}(\mathcal{F}_T)$ be the subgroup of already identified overlap actions generated by the declared color-center and fermion-parity identifications, including any registered continuous compensator used in an overlap test. The subgroup actually removed is

$$K_S = \text{Ker}(G_S \longrightarrow \text{Aut}(\mathcal{A}_T^{\text{phys}})) = G_S \cap \mathcal{O}_T. \quad (20)$$

Because K_S is a kernel—and hence a normal subgroup of G_S —the physical internal remnant is well defined by

$$H_{\text{phys}}(S) = \text{Qualify}(G_S/K_S). \quad (21)$$

The intersection is established from the complete elementary-field action; neither an abstract group order nor a shared name is sufficient.

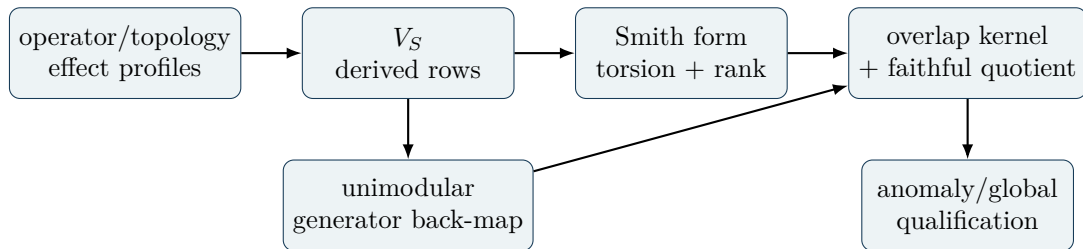


Figure 5: Violation-lattice to Smith form to faithful/anomaly-qualified remnant flow. A Smith invariant without its generator back-map is not accepted as a physical label.

Table 14: The twelve fixed effect sets and their compressed (B, L) transfer rows. The Smith analysis itself is performed on the full primitive lattice $(3B, L_e, L_\mu, L_\tau)$.

Effect set	Constituent effects	Compressed rows ($\Delta B, \Delta L$)	Support
EW	electroweak topological transition	(3, 3)	topological amplitude
W5	Weinberg operator	(0, 2)	generic nonzero
EW + W5	topology + Weinberg	(3, 3), (0, 2)	combined target
BNV6	dimension-six BNV sector	(1, 1)	complete stated sector
EW + BNV6	topology + dimension-six BNV	(3, 3), (1, 1)	combined target
W5 + BNV6	Weinberg + dimension-six BNV	(0, 2), (1, 1)	combined target
EW + W5 + BNV6	topology + Weinberg + BNV6	(3, 3), (0, 2), (1, 1)	combined target
D7-LNV	dimension-seven LNV component	(0, 2)	complete stated component
D7-BLNV	dimension-seven BLNV component	(1, -1)	complete stated component
D9- $\Delta B = 2$	selected dimension-nine sector	(2, 0)	complete selected sector
Majorana	renormalizable M_R	(0, 2)	generic nonzero M_R
Majorana + EW	renormalizable M_R + topology	(0, 2), (3, 3)	combined target

7.2 The twelve fixed effect sets

The twelve sets include the single effects and only those combinations used in the analysis. Family-generic $\Delta L = 2$ support expands to the six symmetric-pair rows, while baryon-number-violating support expands to one row for each lepton family. Exact duplicate rows are then removed before the Smith calculation.

The compressed (B, L) rows are retained as review aids. They reproduce familiar one- or two-charge Smith factors, but they are not the final physical computation. For example, the electroweak row (3, 3) has compressed factor 3, while its primitive four-charge lift (9, 1, 1, 1) has gcd one and therefore no torsion; the three continuous null directions carry the actual selection information. Conversely, the combined electroweak and Weinberg rows span full rank in the primitive lattice and produce raw factor 18, which maps to 6 after baryon normalization and to an independent order-3 class after the fermion-parity overlap.

For every set the exact integer matrices U, D, V satisfy $UV_S V = D$ with $\det U = \pm 1$ and $\det V = \pm 1$. The inverse Smith transformations recover the continuous and torsion generators in the original charge basis, after which their actions are evaluated on the elementary fields. The color-center and fermion-parity intersections are then solved at the level of those field actions and quotiented only when the actions coincide. For the Warsaw-basis $QQQL$ probe, the field multiplicities $Q^3 L_e$ give raw Z_{18} phase exponent 12, while the overlap kernel $H = \langle g^3 \rangle$ has order 6. Since 12 is divisible by 6, the character annihilates H and descends with exponent

$$q_{QQQL} = 12/6 = 2 \pmod{3}. \quad (22)$$

The same derivation gives zero color-center exponent and even fermion parity. This character is evaluated on the local gauge-invariant operator algebra associated with the relevant remnant targets; it does not assert that the $QQQL$ Wilson coefficient is itself nonzero in those effect sets.

Table 15: Smith-normal-form results on the primitive field-charge lattice. “Independent finite” is the finite factor remaining after baryon normalization and faithful color-center/fermion-parity identifications.

Effect set	Cont. rank	Field SNF	Raw torsion	Independent finite	Continuous / recovered generators
EW	3	(1)	none	none	$B - L, L_e - L_\mu, L_\mu - L_\tau$
W5	1	(1, 1, 2)	$\mathbb{Z}_2$	none	B ; lepton parity is not independent
EW + W5	0	(1, 1, 1, 18)	$\mathbb{Z}_{18}$	$\mathbb{Z}_3$	finite class after color and parity quotients
BNV6	1	(1, 1, 1)	none	none	$B - L$
EW + BNV6	1	(1, 1, 1)	none	none	$B - L$
W5 + BNV6	0	(1, 1, 1, 6)	$\mathbb{Z}_6$	none	residual order two is overlap
EW + W5 + BNV6	0	(1, 1, 1, 6)	$\mathbb{Z}_6$	none	residual order two is overlap
D7-LNV	1	(1, 1, 2)	$\mathbb{Z}_2$	none	B ; lepton parity is not independent
D7-BLNV	1	(1, 1, 1)	none	none	$B + L$
D9- $\Delta B = 2$	3	(6)	$\mathbb{Z}_6$	none	L_e, L_μ, L_τ ; baryon parity is absorbed
Majorana	1	(1, 1, 2)	$\mathbb{Z}_2$	none	B ; lepton parity is not independent
Majorana + EW	0	(1, 1, 1, 18)	$\mathbb{Z}_{18}$	$\mathbb{Z}_3$	finite class after color and parity quotients

7.3 Interpretation and safeguards

Several raw $\mathbb{Z}_2$ factors are not new internal matter parities. In the Weinberg-only and Majorana-only targets the lepton-parity element is absorbed, modulo fermion parity, by a continuous baryon phase. In the Weinberg-plus-BNV set the raw factor six becomes order two after the color quotient and is then removed as gauge/fermion-parity overlap. The selected $\Delta B = 2$ set leaves three continuous lepton phases; its baryonic order-two class is similarly absorbed modulo those phases and fermion parity. These statements concern the declared local algebra and effect support, not an arbitrary UV completion.

Two target-distinct full-rank effect sets—electroweak plus Weinberg and electroweak plus renormalizable Majorana—each leave one internal order-three class after the target-local overlap quotient. These are two occurrences on different target instances, not simultaneous factors of one target. Each is classified as an operator-level finite remnant pending a separate global test. Neither is called baryon triality, proton hexality, or a discrete gauge symmetry because global-form and discrete-anomaly completion lie outside the present target.

Theorem 7.1 (Derived violation-lattice theorem). *For each of the twelve fixed effect sets, the violation matrix is the exact deduplicated concatenation of the charge-transfer rows of its constituent effects.*

Proof. The operator-signature analysis fixes the charge-transfer row of each operator or topological effect. Concatenating the rows belonging to a fixed effect set and removing duplicates therefore determines V_S before any Smith reduction or faithful quotient is performed. $\square$

Theorem 7.2 (Smith-normal-form survivor theorem). *Equation (17) has precisely the continuous ranks and raw torsion factors listed in Table 15.*

Proof. For every k , let δ_k be the gcd of all $k \times k$ minors of V_S . The rank is the largest k with $\delta_k \neq 0$, and the invariant factors are $s_k = \delta_k / \delta_{k-1}$. The corresponding invariant factors are reproduced by

Table 16: Physical interpretation of the finite factors after faithful overlap removal and anomaly qualification.

Effect set	Faithful result	Qualification
EW	continuous survivors only	electroweak-null directions from the null-lattice analysis
W5	no independent finite factor	lepton parity absorbed modulo $(-1)^F$ and continuous B
EW + W5	one target-local $\mathbb{Z}_3$ operator remnant	local/EW compatible; global discrete test remains open
BNV6	continuous $B - L$ direction	anomaly qualification depends on the target carrier
EW + BNV6	continuous $B - L$ direction	electroweak compatible; background anomalies treated separately
W5 + BNV6	no independent finite factor	apparent finite factor removed by color/parity overlap
EW + W5 + BNV6	no independent finite factor	apparent finite factor removed by color/parity overlap
D7-LNV	continuous B direction	no independent lepton-parity remnant
D7-BLNV	continuous $B + L$ direction	topology would break it in an EW-extended target
D9- $\Delta B = 2$	three continuous lepton phases	baryon parity absorbed modulo lepton phases and $(-1)^F$
Majorana	continuous B direction	explicit Majorana breaking; global question separate
Majorana + EW	one target-local $\mathbb{Z}_3$ operator remnant	local/EW compatible; global discrete test remains open

explicit unimodular U, D, V transformations, giving

S	(s_i)
EW	(1)
W5 or Majorana or D7-LNV	(1, 1, 2)
EW+W5 or EW+Majorana	(1, 1, 1, 18)
BNV6, EW+BNV6, or D7-BLNV	(1, 1, 1)
W5+BNV6, EW+W5+BNV6	(1, 1, 1, 6)
D9 $\Delta B = 2$	(6).

These values reproduce the ranks and invariant factors shown for all twelve effect sets in Table 15. $\square$

Theorem 7.3 (Faithful-remnant theorem). *Applying the normalization, color-center quotient, fermion-parity identification, and generator back-map gives exactly the independent torsion and faithful status of Tables 15 and 16.*

Proof. The charge-lattice analysis fixes the two lattices and the elementary overlap actions. Unimodular Smith transformations recover each abstract finite factor as an explicit charge combination, whose action can then be evaluated on every field. Equality of the full action vector determines the intersection with the color center and fermion parity, and the subgroup generated by those overlaps determines the physical quotient order. Equation (22) is then a direct character calculation on the resulting quotient rather than an identification by name or order alone. $\square$

Theorem 7.4 (Anomaly-compatible remnant theorem). *A remnant is reported as exact only within*

the anomaly/background scope actually checked; unresolved global conditions remain explicitly stated.

Proof. The anomaly qualification combines the electroweak nullspace result with the local $B - L$ anomaly analysis and the effect target. Electroweak-null directions are distinguished from the minimal-target cubic/gravitational obstruction, and local polynomial cancellation is distinguished from global completion. The two target-distinct order-three occurrences therefore remain explicitly pending the additional global tests needed for a discrete-gauge interpretation. $\square$

Theorem 7.5 (Fixed-set completeness theorem). *The twelve rows of Table 14 are exactly the fixed effect sets used in the remnant analysis.*

Proof. The analysis includes the single-effect sets EW, W5, BNV6, D7-LNV, D7-BLNV, D9- $\Delta B = 2$, and Majorana, together with the stated combinations EW+W5, EW+BNV6, W5+BNV6, EW+W5+BNV6, and Majorana+EW. No remnant statement in the paper uses an effect set outside this list. $\square$

Theorem 7.6 (Canonical remnant identity theorem). *Two remnant presentations receive one canonical identity exactly when their target, effect support, quotient module, back-mapped faithful action, and anomaly/global qualification agree.*

Proof. Smith diagonal entries alone are insufficient: distinct embedded sublattices may have isomorphic abstract quotients. The canonical projection therefore retains set fingerprint, target IDs, source effects, invariant factors, continuous generators, finite generator actions, overlap resolution, and qualification status. Unimodular presentation changes preserve this projection, while a target, support, or faithful-action change does not. $\square$

Summary. Physical remnants are derived in the order effect support $\rightarrow$ violation lattice $\rightarrow$ Smith data $\rightarrow$ generator recovery $\rightarrow$ faithful quotient $\rightarrow$ anomaly/global qualification. Neither a gcd nor a familiar discrete-symmetry label substitutes for this chain.

Part IV

Integrated Closure and Arc Interfaces

8 Canonical typed-result atlas

The finite typed-result theorem is easiest to assess when the classified population is visible. Tables 18–21 therefore list all 34 canonical typed scientific results in reader-facing form. Each row records its result type, target or declared scope, scientific object, and distinguishing conclusion in the same vocabulary used in the body of the paper. The result universe decomposes as

$$\begin{aligned}
 34 = & 31 \text{ target-local results} + 1 \text{ cross-target constraint} \\
 & + 1 \text{ target-family result} + 1 \text{ integrated closure result.}
 \end{aligned}
 \tag{23}$$

The seven scientific-family counts remain

$$(5, 5, 6, 4, 6, 2, 6), \quad (24)$$

for renormalizable origin, electroweak topology, neutrino structure, local $B - L$ qualification, dimension-graded operators, cross-cutting constraints, and remnant closure respectively.

Completeness is not inferred from the displayed count alone. Table 17 states the finite traversal coordinates, target guards, exclusions, and output types that determine the admitted population before the atlas rows are identified.

Table 17: Finite generation and exclusion rules for the 34-result typed atlas. Counts are consequences of the displayed traversal coordinates after the stated target guards are imposed; they are not input cardinalities.

Family	Traversal coordinates	Admission and exclusion rule	Typed output	Rows
Renormalizable origin	minimal, Dirac- ν_R , Majorana- ν_R schemas; B , L_i , total L ; fixed Y_e, Y_ν, M_R strata	Keep charge actions neutral on the complete $d \leq 4$ grammar and stabilizing the fixed tensors. Electroweak, anomaly-background, and higher-dimensional branches are successor schemas.	5 target-local	1–5
Electroweak topology	EW specialization; B , L_i , $B \pm L$, and family differences	Apply the single transfer vector $(3, 1, 1, 1)$. Its non-null aggregate directions and rank-three integer kernel give the five current/transfer results; local EFT branches are excluded.	5 target-local	6–10
Neutrino structure	minimal, Dirac, Weinberg, and Majorana schemas; total- L and family-difference stabilizers	Compute the stabilizer of the fixed tensor set. Generic Y_ν leaves diagonal total L ; generic κ or M_R removes continuous total L . Texture-specific enhancements require a new declared stratum.	6 target-local	11–16
Local $B - L$ qualification	minimal or three- ν_R carrier; mixed-gauge, gravitational/cubic, and global-completion probes	Separate mixed-gauge cancellation, the minimal gravitational/cubic obstruction, three- ν_R local cancellation, and the unresolved global frontier. Local cancellation is not promoted to global gaugeability.	4 target-local	17–20
Dimension-graded operators	$d = 5, 6, 7, 9$; minimal or ν_R carrier; exact $(\Delta B, \Delta L)$ support	Admit only the stated complete or selected-complete sectors: Weinberg; minimal and ν SMEFT $d = 6$ BNV; the two corrected $d = 7$ components; and the selected complete $d = 9$, $\Delta B = 2$ sector.	6 target-local	21–26
Cross-cutting constraints	local operator dimension/charge parity; EW zero-mode vertex	Apply the parity theorem across the registered family of local-operator targets. Retain the dimension-18 EW vertex as a target-local topology-dependent amplitude, not a Wilson-operator class.	1 constraint + 1 target-local	27–28
Remnant closure	12 effect-set instances; violation rows; Smith data; back-map; overlap kernel; anomaly/global qualification	Construct all twelve admitted $T(S)$ target instances and their target-local remnant calculations. The publication atlas records six typed remnant results: four target-local entries, one indexed $d = 7$ target-family synthesis, and one all-registry closure result. No equivalence between distinct target instances is inferred merely from an isomorphic remnant group.	4 local + 1 family + 1 closure	29–34
Total		Every admitted result constructor lands in one typed atlas row; every atlas row has a generated antecedent; every excluded branch changes a declared coordinate or lies outside the fixed source sectors.	34 = 31 + 1 + 1 + 1	1–34

Table 18: Canonical typed results 1–10: renormalizable origin and electroweak topology.

No.	Result type	Target or declared scope	Scientific object	Distinguishing physical content	Canonical conclusion
1	Target-local	Minimal renormalizable SM	baryon action	complete $d \leq 4$ grammar; physical $B = 1/3$ on quarks	faithful accidental $U(1)_B$ before topology
2	Target-local	Minimal renormalizable SM	family lepton actions	charged-lepton tensor only; no Y_ν , M_R , or κ	three independent family phases
3	Target-local	Dirac- ν_R target	baryon action	three singlet neutrinos, $M_R = 0$	same faithful baryon action as the minimal carrier
4	Target-local	Dirac- ν_R target	total lepton action	generic Y_ν with $M_R = 0$	continuous total $U(1)_L$; family differences need not survive
5	Target-local	Majorana- ν_R target	total lepton direction	generic nonzero M_R	continuous L is broken by $\Delta L = 2$
6	Target-local	Minimal SM with EW topology	baryon current/transfer	$Q_{EW} \in \mathbb{Z}$	$\Delta B = 3Q_{EW}$; current anomalous across topology
7	Target-local	Minimal SM with EW topology	family lepton currents/transfer	one lepton zero-mode per family	$\Delta L_i = Q_{EW}$; family sum changes
8	Target-local	Minimal SM with EW topology	$B + L$ direction	physical transfer row $(3, 1, 1, 1)$	$\Delta(B + L) = 6Q_{EW}$
9	Target-local	Minimal SM with EW topology	$B - L$ direction	integral kernel of electroweak transfer	continuous electroweak-null direction
10	Target-local	Minimal SM with EW topology	family-difference directions	same transfer in all three lepton families	$L_e - L_\mu$ and $L_\mu - L_\tau$ lie in the EW null lattice

Table 19: Canonical typed results 11–20: neutrino targets and the qualified $B - L$ boundary.

No.	Result type	Target or declared scope	Scientific object	Distinguishing physical content	Canonical conclusion
11	Target-local	Minimal massless-neutrino target	family lepton stabilizer	Y_e only	three target-local family $U(1)$ actions
12	Target-local	Dirac- ν_R target	total lepton stabilizer	generic Y_e, Y_ν with $M_R = 0$	one diagonal continuous $U(1)_L$
13	Target-local	Dirac- ν_R target	family-difference directions	generic misalignment of Y_e and Y_ν	family differences are not generic survivors
14	Target-local	Weinberg target	total lepton direction	nonzero dimension-five κ	$\Delta L = 2$; no continuous total L
15	Target-local	Weinberg target	family-difference directions	generic symmetric κ	generic family differences are not continuous survivors
16	Target-local	Majorana- ν_R target	total lepton direction	generic full-rank M_R	$\Delta L = 2$; no continuous total L
17	Target-local	Minimal $B - L$ background target	mixed SM gauge anomalies	$SU(3)_c, SU(2)_L, U(1)_Y$ probes	mixed dynamical-gauge coefficients vanish
18	Target-local	Minimal $B - L$ background target	gravitational and cubic background anomalies	no right-handed neutrino	local polynomial obstruction remains
19	Target-local	Three- ν_R $B - L$ background target	gravitational and cubic background anomalies	one singlet neutrino per family	local polynomial deficits cancel
20	Target-local	Three- ν_R $B - L$ background target	gauging/global completion	local cancellation established, global form not supplied	global gaugeability remains a successor question, not a present theorem

Table 20: Canonical typed results 21–28: dimension-graded operator sectors and cross-cutting constraints.

No.	Result type	Target or declared scope	Scientific object	Distinguishing physical content	Canonical conclusion
21	Target-local	Weinberg EFT	dimension-five class	unique local $d = 5$ operator; generic nonzero support	$(\Delta B, \Delta L) = (0, 2)$
22	Target-local	Minimal SMEFT through $d = 6$	BNV dimension-six sector	four complete minimal BNV types	$(\Delta B, \Delta L) = (1, 1)$; $B - L$ operator-null
23	Target-local	ν SMEFT through $d = 6$	right-handed-neutrino BNV sector	two additional singlet-neutrino types	same charge direction, with Majorana-stratum qualification retained
24	Target-local	Minimal SMEFT at $d = 7$	LNV component	corrected complete twelve-type sector	$(\Delta B, \Delta L) = (0, 2)$
25	Target-local	Minimal SMEFT at $d = 7$	BLNV component	corrected complete six-type sector	$(\Delta B, \Delta L) = (1, -1)$
26	Target-local	Selected complete $d = 9$ sector	$\Delta B = 2$ operators	complete five-type sector plus Hermitian conjugates	$(\Delta B, \Delta L) = (2, 0)$
27	Constraint	Registered family of local-operator targets	dimension/ $B - L$ -parity constraint	cross-target imported parity predicate, applied separately to each admitted local-operator target	the parity relation holds separately on every admitted local-operator target
28	Target-local	Electroweak topological target	't Hooft zero-mode vertex	nine quark-doublet plus three lepton-doublet insertions	dimension 18, $(\Delta B, \Delta L) = (3, 3)$, topological rather than Wilson-operator support

Table 21: Canonical typed results 29–34: remnant closure. These comprise four target-local remnant results, one target-family synthesis, and one integrated closure result. The 12 effect sets are tabulated separately in Table 14.

No.	Result type	Target or declared scope	Scientific object	Faithful quotient or component data	Canonical conclusion
29	Target-local	Electroweak topology only	EW null-lattice remnant	rank-three continuous kernel; no independent finite torsion	$B - L$ and two family differences survive at the topology level
30	Target-local	Weinberg operator only	Weinberg remnant	one continuous baryon direction; apparent lepton parity absorbed modulo $(-1)^F$	continuous $U(1)_B$, no independent finite factor
31	Target-local	Dimension-six BNV only	BNV6 remnant	one continuous null direction	continuous $B - L$ on the declared operator target
32	Target family	Indexed pair $\{T_{d=7,\text{LNV}}, T_{d=7,\text{BLNV}}\}$	dimension-seven remnant-family synthesis	two target-local components with different continuous null directions	B survives on the LNV target; $B + L$ survives on the restricted BLNV target
33	Target-local	Selected $d = 9$, $\Delta B = 2$ sector	baryon-two remnant	three lepton phases; apparent baryon parity absorbed by existing overlaps	no independent finite factor
34	Closure	Registry $\{T(S_i)\}_{i=1}^{12}$	integrated remnant closure	all twelve component calculations retain generator back-maps, overlap quotients, and local anomaly qualifications	the component conclusions are exactly those of Tables 15 and 16; two target-distinct instances leave order-three operator-level remnants pending SYM-9 completion

The atlas makes three facts explicit. First, the number 34 is exposed directly in the paper: every typed result appears above as a concrete scientific object. Second, the 31 target-local entries are compared only under the target-preserving relation $\sim_T$; the constraint, target-family, and closure entries retain their declared domains and component projections through Π_7 . Third, the same familiar charge can occur in several target-local rows without duplication because the target or physical disposition is different. Baryon number as a renormalizable accidental action, baryon-current transfer across electroweak topology, and baryon violation by a dimension-six operator are different target-local statements connected by explicit ancestry, not three presentations of one record.

9 Integrated closure theorem package

The preceding parts now support the endpoint claim in the same architectural form used elsewhere in the Symmetry Arc: physical construction first, target-local canonicalization where applicable, typed integration second, and then an explicit successor interface. The scientific content can be summarized in seven principal results.

Table 22: Principal SYM-7 results and their scientific content.

Result	Content	Primary sections
Accidental-origin theorem	Complete renormalizable grammar yields baryon number and the target-appropriate lepton stabilizer as derived accidental actions	3
$B + L$ anomaly/topology theorem	Action, current anomaly, perturbative selection, and topological transfer remain distinct; unit EW topology has physical transfer $(3, 1, 1, 1)$	4
Qualified $B - L$ theorem	$B - L$ is electroweak-null, but its local background-anomaly and global-gauging status depends on neutrino content and target data	5
Weinberg-operator theorem	Nonzero dimension-five Weinberg support gives $\Delta L = 2$ and changes both total- and family-lepton symmetry status	5–6
Baryon/lepton violation by dimension	The registered $d = 6$, corrected $d = 7$, selected complete $d = 9$, and EW topological sectors have exact target-qualified charge signatures	6
Discrete-remnant theorem	Each finite or continuous survivor is derived from the violation lattice, Smith data, faithful field action, overlap quotient, and anomaly/global qualification	7
No primitive promotion	None of the resulting accidental actions or remnants becomes a primitive carrier principle merely because it survives the declared constraints	9

Theorem 9.1 (Finite typed-result completeness). *Relative to the 13 registered target schemas of Table 2, the target instances obtained by supplying the twelve fixed effect sets of Table 14, and the stated complete or selected-complete operator sectors, the declared SYM-7 generation protocol produces exactly 34 canonical typed scientific results:*

31 target-local results $\sqcup$ 1 cross-target constraint $\sqcup$ 1 target-family result $\sqcup$ 1 integrated closure result.

Every admitted generated branch has exactly one typed atlas representative, every atlas entry has a

generated antecedent, and no untyped result remains inside the declared finite SYM-7 domain.

Proof. Table 17 gives the finite generator family by family. In the renormalizable, topology, neutrino, local- $B - L$, operator, cross-cutting, and remnant sectors, respectively, its admission rules produce (5, 5, 6, 4, 6, 2, 6) compatible branches. The same table states the corresponding exclusions: a rejected branch either changes a target coordinate, requires an operator or effect sector not admitted by the theorem, or belongs to a texture, global-form, RG, or other successor domain. Tables 18–21 exhibit one typed result for every admitted branch. Rows 1–26, 28–31, and 33 are target-local and are canonicalized by π_T under $\sim_T$. Row 27 is a cross-target constraint, row 32 is an indexed target-family result, and row 34 is an integrated closure result; their declared domains and component projections are retained by Π_7 . Thus every generated branch lands in exactly one typed atlas row, every atlas row has a generated antecedent, and the seven family counts sum to 34 with type distribution $31 + 1 + 1 + 1$ and no additional compatible branch in the declared finite domain. $\square$

Theorem 9.2 (Typed canonical uniqueness). *The 34 entries in the SYM-7 atlas are pairwise distinct under the type-indexed canonical projection Π_7 . The 31 target-local entries represent 31 distinct target-preserving $\sim_T$ classes. The cross-target constraint, target-family result, and integrated closure result are distinguished by their declared type, scope, component projections, and conclusion.*

Proof. For target-local rows, admissible representative transformations preserve π_T , while every pair of distinct local rows differs in at least one target-local class-defining coordinate: target, faithful action, operator support, perturbative status, anomaly qualification, topology, neutrino stratum, dimension/charge signature, violation lattice, remnant, or observable class. Hence the 31 local rows represent 31 distinct $\sim_T$ classes. Rows 27, 32, and 34 carry distinct type tags—constraint, target family, and closure—and Π_7 also retains their target-family domains, ordered component projections, conclusions, and exact nonclaims. No higher typed result can therefore collapse into a local result or into another higher result. The 34 atlas entries are pairwise distinct under $\equiv_7$. $\square$

Theorem 9.3 (Integrated accidental-charge classification). *Within the registered target schemas and their declared instances in Table 2, baryon and lepton actions, their perturbative and topological selection rules, neutrino-tensor breaking, local anomaly qualifications, dimension-graded operator violations, and faithful continuous or finite remnants are jointly classified without changing target in the course of an inference.*

Proof. The renormalizable-origin results establish the accidental charge directions. The electroweak analysis separates current anomaly, perturbative selection, and topological transfer. The neutrino analysis distinguishes the four neutrino targets and separates electroweak-null $B - L$ from its local cubic/gravitational obstruction and from the stronger question of global gaugeability. The operator analysis determines the exact registered charge signatures, and the remnant analysis constructs the violation lattices, Smith factors, faithful quotients, and anomaly-qualified survivors. Because every cross-section statement is indexed by the target data of Equation (1), these results assemble without identifying inequivalent carrier, operator, flavor, or quantum sectors. $\square$

Theorem 9.4 (No primitive promotion). *No accidental action, perturbative selection rule, observed longevity, zero-support stratum, or derived finite remnant in this classification is thereby promoted to a primitive symmetry principle.*

Proof. Baryon invariance in the minimal renormalizable theory follows from monomial exhaustion; the electroweak qualification follows from the separation of action, current, and topological transfer; neutrino statements follow from explicit tensor strata; and each finite remnant is derived from a specified violation lattice followed by faithful quotienting. A vanishing coefficient defines a different support stratum rather than a fundamental prohibition. The surviving symmetries are therefore residues of the specified physical constraints, not additional laws inserted above them. $\square$

10 Downstream interfaces and successor discipline

Closure at SYM-7 does not mean that every stronger symmetry question has been answered. The arc instead exports three explicit successor packets, summarized in Table 23. This is the same successor discipline used by SYM-3 and SYM-4: a downstream refinement inherits the predecessor target and result, then declares any changed coordinate rather than silently rewriting the earlier theorem. [4, 5]

Table 23: Downstream interfaces from SYM-7.

Arc	Receives from SYM-7	Successor burden	What SYM-7 does not claim
SYM-9	charge lattices, gauge overlaps, interpreted remnant generators, local anomaly/topology status, target fingerprints	faithful global form, line/monopole sectors, discrete anomalies, special- θ , higher-group and non-invertible completion	no present promotion of either target-distinct order-three operator remnant to an exact discrete gauge symmetry
SYM-10	operator classes, charge signatures, coefficient-support strata, EFT order, remnant invariants	RG and threshold transport while preserving the physical charge sector	no numerical Wilson coefficients, matching scales, or threshold data
SYM-11	the 34-result typed atlas, including 31 target-local classes and three higher typed results; 13-schema target family; 12 effect-set target instances; ancestry and nonclaims	whole-arc candidate exhaustion and integrated symmetry closure	no claim that SYM-7 alone exhausts every symmetry carrier in the Standard Model program

The SYM-9 interface is particularly important. For each of the two target-distinct effect sets EW+Weinberg and Majorana+EW, the local faithful quotient leaves one order-three operator-level class on that target instance. The two occurrences are not simultaneous factors of a single target. Each statement is complete at its present local-algebra target. Whether the class extends to an exact discrete gauge symmetry requires the global-form and discrete-anomaly information deliberately assigned downstream. Conversely, later global analysis may refine the interpretation but must not retroactively change the local Smith and faithful-action calculation proved here.

The SYM-10 interface carries the opposite sort of information: operator identity, support, and charge signature are fixed here, while their scale dependence is not. A basis or scheme transformation may be a presentation change only when the invariant charge content is transported faithfully. Numerical running and matching remain target data of the RG paper rather than hidden inputs to the present classification.

11 Exact scope, falsifiability, and interpretation

The finite scope makes the principal failure modes concrete. The classification would require revision if one found, within the same declared target, a missing gauge-invariant $d \leq 4$ monomial carrying nonzero claimed accidental charge; a different electroweak transfer vector or integer null lattice; an incorrect dimension or $(\Delta B, \Delta L)$ assignment in one of the declared complete sectors; a different Smith invariant factor for a fixed effect set; a finite generator whose full field action is trivial or already absorbed by an omitted overlap; or a local anomaly qualification inconsistent with the stated background and carrier content.

By contrast, adding fields, changing coefficient support, turning on a Majorana tensor, admitting a new operator sector, or changing global-form data changes the target. Such a successor may be physically important, but it is not a counterexample to a theorem whose antecedent explicitly fixes the predecessor target. This is the same fixed-domain discipline used throughout the arc. [1, 2, 4]

Several familiar statements can now be read as different projections of the integrated classification. In the minimal renormalizable target, B is accidental because all admitted $d \leq 4$ monomials are neutral. In the nontrivial electroweak sector, the integrated transfer removes $B + L$ while leaving $B - L$ and two family differences in the integer null lattice. Generic Dirac-neutrino Yukawa support preserves total lepton number but removes generic family differences; turning on M_R or the Weinberg tensor removes continuous total L through a $\Delta L = 2$ effect. Higher-dimensional BNV and LNV operators then select further violation directions. Finite factors are retained only after generator recovery, comparison with color-center and fermion-parity overlaps, and anomaly/global qualification.

12 Conclusion

Accidental symmetry is a derived property of a specified physical target. Once field content, operator support, flavor stratum, background content, and quantum sector are fixed, the apparent tensions among classical conservation laws, anomalous currents, topological selection rules, neutrino-mass breaking, higher-dimensional operators, and discrete remnants become one target-indexed problem in faithful charge actions and violation lattices.

For the finite family studied here, the minimal renormalizable theory yields accidental baryon number and three independent family lepton phases; electroweak topology has transfer vector $(3, 1, 1, 1)$ and a rank-three null lattice; the four neutrino targets distinguish massless, Dirac, renormalizable-Majorana, and Weinberg breaking; dimensions five, six, seven, and the selected dimension-nine sector supply the registered charge-violation directions; and twelve fixed effect sets determine the continuous and finite remnants after Smith reduction and faithful quotienting. The complete reader-facing atlas contains 34 canonical typed scientific results across the declared 13-schema domain and its effect-set-instantiated remnant targets: 31 target-local classes, one cross-target constraint, one target-family synthesis, and one integrated closure result.

The broader structural conclusion is the one required by the Symmetry Arc. None of these symmetries is promoted because its name is familiar, its violation is small, or an abstract quotient is nontrivial. Its physical standing is whatever survives the fixed carrier, grammar, tensor, quantum-sector, support, and faithful-action constraints. SYM-7 therefore closes accidental baryon/lepton charge at its declared operator and topology scope while handing the genuinely stronger global, RG, and whole-arc questions forward without conflation.

A Literature and arc-source register

The classification separates imported source theorems from local specializations. Table 24 records the principal external inputs that carry completeness or anomaly information. The paper then specializes those inputs to the fixed targets of Table 2; it does not re-label an imported basis theorem as a new derivation.

Table 24: Principal external source interfaces.

Source	Imported result	Local use	Nonclaim
't Hooft [9, 10]	electroweak zero-mode selection and fermion-number violation	three-generation transfer vector and dimension-18 topological vertex	no transition rate or thermal history
Klinkhamer–Manton [11]	electroweak sphaleron saddle	quantum-sector context for the topology interface	no cosmological rate calculation
Weinberg [12]	unique dimension-five LNV class	$\Delta L = 2$ charge vector and Weinberg target	no ultraviolet completion or coefficient prediction
Warsaw basis [14]	four minimal dimension-six BNV operator types	complete minimal $d = 6$ BNV charge sector	no proton-decay rates or Wilson coefficients
Alonso et al. [15]	right-handed-neutrino BNV extension	two additional ν_R -containing $d = 6$ types	no fixed Majorana stratum or numerical RG evolution
Liao–Ma $d = 7$ [16]	corrected twelve LNV plus six BLNV types	complete charge-relevant $d = 7$ split	no arbitrary-dimension completion
Kobach [18]	operator-dimension/ $B - L$ parity relation	cross-check across registered local sectors	no statement about nonperturbative topological amplitudes
Liao–Ma $d = 9$ [17]	complete dimension-nine basis	selected complete $\Delta B = 2$, $\Delta L = 0$ sector	no claim about all dimension-nine phenomenology
Putrov–Wang [13]	sterile-count/gravitational-anomaly boundary	local $B - L$ background qualification	no non-invertible or global completion

The internal arc sources play a different role. SYM-0 through SYM-6 fix the status, faithful-action, anomaly, discrete, flavor, and chiral interfaces summarized in Table 1. Their use is target-preserving: SYM-7 inherits the relevant result and then adds accidental-charge coordinates rather than modifying the predecessor statement. [1–7]

B Operator-basis and convention notes

The dimension-five class follows Weinberg [12]. The four minimal dimension-six baryon-number-violating operator types follow the Warsaw-basis classification [14]; the two additional right-handed-neutrino types follow the ν SMEFT extension [15]. The dimension-seven split into twelve lepton-number-violating and six baryon-and-lepton-number-violating types follows Liao and Ma [16]. The selected complete $d = 9$, $\Delta B = 2$, $\Delta L = 0$ sector follows Liao and Ma [17]. The dimension/charge parity relation is taken from Kobach [18].

Throughout the charge-lattice calculation, the primitive integral basis is $(3B, L_e, L_\mu, L_\tau)$ and the physical basis is (B, L_e, L_μ, L_τ) . Right-handed fermions may be represented by their left-handed charge-conjugate Weyl fields; anomaly signs and operator charge assignments are kept in one fixed convention within each calculation. The electroweak transfer vector $(3, 1, 1, 1)$ is always quoted in physical charges, while $(9, 1, 1, 1)$ denotes the corresponding primitive integral lift.

The phrase “complete sector” is used only where an imported basis result establishes completeness for the stated field content and order. The dimension-nine discussion is intentionally a selected complete $\Delta B = 2$ sector rather than a claim that every dimension-nine operator is relevant to the present accidental-charge question. Likewise, the dimension-18 electroweak ’t Hooft vertex is retained as a topological amplitude and is not inserted into the SMEFT Wilson-operator basis merely because it has a canonical dimension.

C Computational companion and exact arithmetic

The project accompanying this manuscript preserves the technical SYM-7 source archive containing the machine-readable target, field, charge, anomaly, operator, effect-set, Smith, remnant, and canonical-record data used for the finite calculations. It also contains exact-arithmetic scripts for charge signatures, null lattices, Smith normal forms, generator back-maps, overlap quotients, and target-indexed comparisons. These materials are supplementary verification resources; the scientific argument and the complete 34-result typed population is presented in reader-facing form in the article itself.

The computational division of labor is intentionally narrow. Published operator-basis completeness, instanton zero-mode structure, anomaly coefficients, and related field-theoretic inputs remain literature results cited in the text. The companion calculations check the finite consequences used here. In particular, the integral charge-basis maps are checked on the full lattice in both directions, and each Smith calculation verifies

$$UV_S V = D \tag{25}$$

with unimodular U and V before any finite factor is assigned a physical interpretation. The generator is then mapped back to the elementary fields and compared with gauge-center and fermion-parity overlaps.

For publication purposes, the technical source archive is kept under **verification/** rather than typeset as raw ledgers. This separation is deliberate: readers can reproduce the finite arithmetic without forcing the article to carry development logs, internal identifiers, or machine serialization. The root `README.md` explains the publication build and the location of the preserved verification materials.

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